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Math 208 Major Exam II Term 243

Original exam date: 20 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = \left(A + Bx + Cx^2\right)\cos 2x + \left(D + Ex + Fx^2\right)\sin(2x)$$

is

(a) $y^{(6)} - 12y^{(4)} + 48y'' + 64y = 0$

(b) $y^{(6)} - 12y^{(4)} - 48y'' + 64y = 0$

(c) $y^{(6)} + 12y^{(4)} + 48y'' - 64y = 0$

(d) $y^{(6)} + 12y^{(4)} + 48y'' + 64y = 0$

(e) $y^{(6)} + 12y^{(4)} - 48y'' + 64y = 0$

2. If $y_p = A + Bxe^x + Cx^2e^x$ is a particular solution of the differential equation

$$y'' + 2y' - 3y = 1 + xe^x,$$

then $9A + 16B =$

(a) 0

(b) -3

(c) -4

(d) 3

(e) 4

3. The general solution of the differential equation

$$y^{(4)} + y^{(3)} - 3y'' - 5y' - 2y = 0$$

is

(a) $y(x) = c_1e^{2x} + c_2e^x + (c_3 + c_4x)\,e^{-x}$

(b) $y(x) = c_1e^{2x} + \left(c_2 + c_3x + c_4x^2\right)e^x$

(c) $y(x) = c_1e^{-2x} + \left(c_2 + c_3x + c_4x^2\right)e^x$

(d) $y(x) = c_1e^{-2x} + c_2e^x + (c_3 + c_4x)\,e^{-x}$

(e) $y(x) = c_1e^{2x} + \left(c_2 + c_3x + c_4x^2\right)e^{-x}$

4. An eigenvector associated with the eigenvalue $\lambda = 5$ of the matrix

$$A = \begin{bmatrix} 9 & -10 \\ 2 & 0 \end{bmatrix}$$

is $\begin{bmatrix} a \\ 2 \end{bmatrix}$ where $a =$

(a) -5

(b) 4

(c) 5

(d) 0

(e) -4

5. Consider the subspace S of $\mathbb{R}^4$ defined by

$$S = \{(x, y, z, w) \mid x + 8z = y + 7w = 0\}.$$

A basis of S consists of the vectors

(a) $\mathbf{v}_1 = (-8, 0, 1, 0)$ and $\mathbf{v}_2 = (0, -7, 0, 1)$

(b) $\mathbf{v}_1 = (-8, 0, 1, 0)$ and $\mathbf{v}_2 = (0, -7, 0, -1)$

(c) $\mathbf{v}_1 = (-8, 0, -1, 0)$ and $\mathbf{v}_2 = (0, -7, 0, 1)$

(d) $\mathbf{v}_1 = (-8, 0, 1, 0)$ and $\mathbf{v}_2 = (0, 7, 0, 1)$

(e) $\mathbf{v}_1 = (8, 0, 1, 0)$ and $\mathbf{v}_2 = (0, -7, 0, 1)$

6. The rank of the matrix

$$A = \begin{bmatrix} 3 & 2 & 4 & 1 \\ 2 & 1 & 3 & 2 \\ 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{bmatrix}$$

is

(a) 1

(b) 2

(c) 5

(d) 4

(e) 3

7. Given that $y = \cos(2x)$ is a solution of the differential equation

$$6y^{(4)} + 5y^{(3)} + 25y'' + 20y' + 4y = 0.$$

The general solution of the differential equation is

(a) $y(x) = c_1e^{-x} + c_2e^{-2x} + c_3\cos(2x) + c_4\sin(2x)$

(b) $y(x) = c_1e^{-x/2} + c_2e^{-x/3} + c_3\cos(2x) + c_4\sin(2x)$

(c) $y(x) = c_1e^{-x/2} + c_2e^{-x/4} + c_3\cos(2x) + c_4\sin(2x)$

(d) $y(x) = c_1e^{-x/2} + c_2e^{x} + c_3\cos(2x) + c_4\sin(2x)$

(e) $y(x) = c_1e^{2x} + c_2e^{-x} + c_3\cos(2x) + c_4\sin(2x)$

8. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$y^{(5)} - y' = (1 + 2x)e^{-x} + 3$$

is given by $y_p(x) =$

(a) $Ax^2 + (B + Cx)\,e^{-x}$

(b) $A + (B + Cx)\,e^{-x}$

(c) $Ax + \left(Bx^2 + Cx^3\right)e^{-x}$

(d) $Ax^2 + \left(Bx^2 + Cx^3\right)e^{-x}$

(e) $Ax + \left(Bx + Cx^2\right)e^{-x}$

9. The characteristic polynomial of the matrix

$$\begin{bmatrix} 1 & 0 & -1 \\ -2 & 3 & -1 \\ -6 & 6 & 0 \end{bmatrix}$$

is $p(\lambda) =$

(a) $-\lambda^3 + 8\lambda^2 - 3\lambda$

(b) $-\lambda^3 + 6\lambda^2 - 3\lambda$

(c) $-\lambda^3 + 4\lambda^2 - 3\lambda$

(d) $-\lambda^3 + 4\lambda^2 + 3\lambda$

(e) $-\lambda^3 + 4\lambda^2 - 2\lambda$

10. If the solution space of the system

$$\begin{aligned} x_1 + 3x_2 - 4x_3 - 8x_4 + 6x_5 &= 0 \\ x_1 + 2x_3 + x_4 + 3x_5 &= 0 \\ 2x_1 + 7x_2 - 10x_3 - 19x_4 + 13x_5 &= 0 \end{aligned}$$

consists of all linear combinations of the three vectors $v_1 = (\alpha, \beta, 1, 0, 0)$, $v_2 = (a, b, 0, 1, 0)$ and $v_3 = (m, n, 0, 0, 1)$, then $\alpha + \beta + a + b + m + n =$

(a) -2

(b) -3

(c) 1

(d) 3

(e) 4

11. A particular solution of the differential equation

$$y'' + 9y = 2\sec(3x)$$

is given by $y_p(x) =$

(a) $\frac{2}{9}\left[x\sin(3x) + \cos(3x)\ln|\cos(3x)|\right]$

(b) $\frac{2}{9}\left[3x\sin(3x) + \cos(3x)\ln|\cos(3x)|\right]$

(c) $\frac{2}{9}\left[3x\cos(3x) + x\sin(3x)\right]$

(d) $\frac{2}{9}\left[3x^2\sin(3x) + 2\cos(3x)\ln|\cos(3x)|\right]$

(e) $\frac{2}{9}\left[3x\sin(3x) + 3x\cos(3x)\right]$

12. If the matrix $A = \begin{bmatrix} 5 & -4 \\ 3 & -2 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 4 \\ -1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

13. If $W(x)$ is the Wronskian of the functions

$$f(x) = x, \quad g(x) = \cos(\ln x), \quad h(x) = \sin(\ln x), \qquad x > 0,$$

then $W(x) =$

(a) $\dfrac{3}{x}$

(b) $\dfrac{2}{x}$

(c) $\dfrac{2}{x^2}$

(d) $\dfrac{1}{x^3}$

(e) $\dfrac{3}{x^2}$

14. If $y(x)$ is the solution of the initial-value problem

$$y'' - 10y' + 25y = 0; \quad y(0) = 3,\ y'(0) = 13,$$

then $y(1) =$

(a) $4e^5$

(b) e^5

(c) $2e^5$

(d) $3e^5$

(e) 0

15. If the characteristic polynomial of the matrix

$$A = \begin{bmatrix} 3 & 6 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is $p(\lambda) = -(\lambda - 1)^2(\lambda - 3)$, then a basis for the eigenspace of $\lambda = 1$ is

$$V_1 = \begin{bmatrix} \alpha \\ 1 \\ 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} \beta \\ 0 \\ 1 \end{bmatrix},$$

where $\alpha + \beta =$

(a) -3

(b) -2

(c) 5

(d) -4

(e) 3

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$r = \pm 2i$ triple: $(r^2+4)^3 = r^6 + 12r^4 + 48r^2 + 64$.
2	**(c)**	$A = -\frac{1}{3}$; $(4B + 2C + 8Cx)e^x = xe^x$ gives $B = -\frac{1}{16}$: $9A + 16B = -4$.
3	**(e)**	$r^4 + r^3 - 3r^2 - 5r - 2 = (r+1)^3(r-2)$: $c_1e^{2x} + (c_2 + c_3x + c_4x^2)e^{-x}$.
4	**(c)**	$(A - 5I)\mathbf{v} = \mathbf{0}$: $2a - 10 = 0$, so $a = 5$.
5	**(a)**	$x = -8z$, $y = -7w$: $z(-8, 0, 1, 0) + w(0, -7, 0, 1)$.
6	**(e)**	Row reduce: pivots in columns 1, 2, 4 only, so $\operatorname{rank}(A) = 3$.
7	**(b)**	$(r^2+4)(2r+1)(3r+1)$: roots $\pm 2i, -\frac{1}{2}, -\frac{1}{3}$.
8	**(e)**	$r = 0, \pm 1, \pm i$ (simple): $y_p = Ax + (Bx + Cx^2)e^{-x}$.
9	**(c)**	Row-1 expansion: $(1-\lambda)(\lambda^2 - 3\lambda) = -\lambda^3 + 4\lambda^2 - 3\lambda$.
10	**(a)**	RREF: $x_1 = -2x_3 - x_4 - 3x_5$, $x_2 = 2x_3 + 3x_4 - x_5$; sum $= 0 + 2 - 4 = -2$.
11	**(b)**	VoP, $W = 3$: $u_1 = \frac{2}{9}\ln\lvert\cos 3x\rvert$, $u_2 = \frac{2}{3}x$.
12	**(e)**	$\lambda = 1$: $(1, 1)$; $\lambda = 2$: $(4, 3)$. $P = [\,(1,1)\ (4,3)\,]$, $D = \operatorname{diag}(1, 2)$.
13	**(c)**	Expand along column 1: $W = x \cdot \frac{1}{x^3} + \frac{1}{x^2} = \frac{2}{x^2}$.
14	**(b)**	$(r-5)^2 = 0$: $y = (3 - 2x)e^{5x}$, so $y(1) = e^5$.
15	**(b)**	$A - I$: $x_1 = -3x_2 + x_3$, so $\alpha = -3$, $\beta = 1$, sum -2.